Drawing – making field sketches and specimen diagrams

Drawing helps you to develop your skills of observation – if you understand something you should be able to explain it in a diagram or drawing.

You need to look at a specimen or a structure very carefully if you are going to be able to make an accurate drawing. Labelling your drawing makes you think about the important components of your drawing and their exact position. If your observations are poor then your drawing will show this.

What is the difference between a drawing, a diagram and a figure?

A drawing is an accurate representation of a specimen or feature like a cliff – it requires only a good eye, some drawing skill but no previous knowledge about the object

A diagram needs to be accurate in its general proportions but can be very stylized showing only the most important features, so you need some knowledge of what to include and what details can be left out.

Diagrams in a report or other scientific literature are often called figures.

Specimen diagrams

The aim is to produce a stylized representation of the specimen indicating the main surface features. The use of shading should in general be avoided. Your diagram should be fully labelled and notes added where relevant. The main problem will be keeping the different parts in proportion to each other. In a diagram you can merge features seen on different specimens to produce your diagram – use the best bits.

Planning - Decide what to draw – what is the purpose of the drawing, what details need to be included, what magnification or reduction (your scale) is needed.

When you have decided these things then your drawing needs to be as large as can fit in the space you have available always leaving room for the labels.

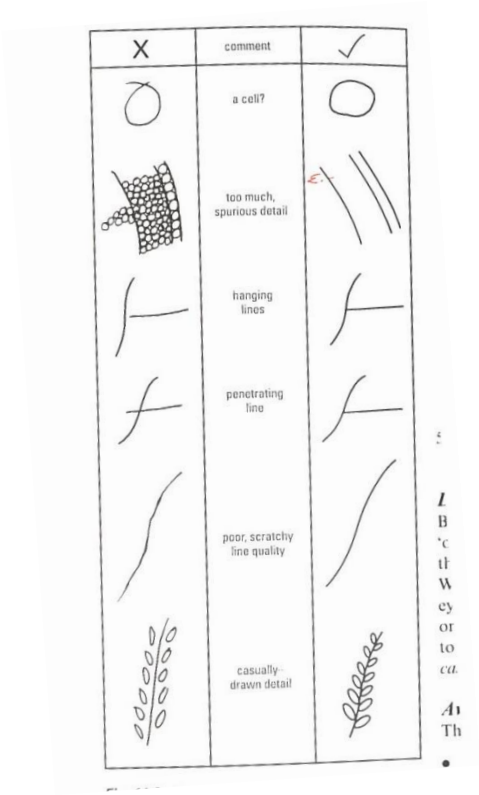
Materials – most drawings are done in pencil – a 2H is best and you need to sharpen your pencil regularly to keep the point sharp. You will need a good quality eraser. If you need to produce a figure for a piece of work then you should produce a line drawing on your digital device (tablet or PC) from your original pencil drawing.

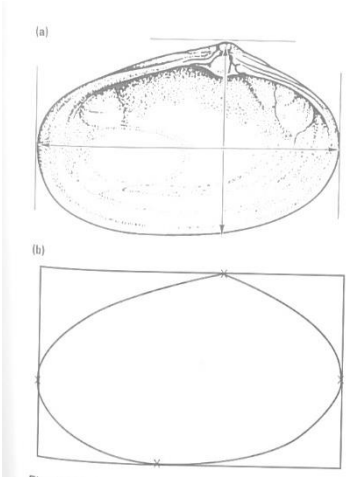
Construction advice –

1. Draw a faint rectangle to show the drawing boundary
2. Draw faint construction lines to get the correct proportions and outlines - these will be erased when the drawing is complete. Use a ruler to work out the dimensions of your object and then using these dimensions draw a scaled frame to allow further reference points to be located
3. Draw the main outlines and then when you are satisfied go over the lines with a sharp pencil. Use firm, continuous lines and make sure that junctions between lines are properly drawn. Always draw what you see not what you think you should see and ask if you can't see the structures or features you are supposed to be drawing.
4. Complete your drawing by adding the labels. Use either radiating or horizontal lines, the lines should not cross. Labels should be written neatly in one orientation (horizontal). Notes

can be added below the labels to show what you have seen or have been pointed out to you by your demonstrator or lecturer.

5. Add a scale or magnification factor, a title, and a legend





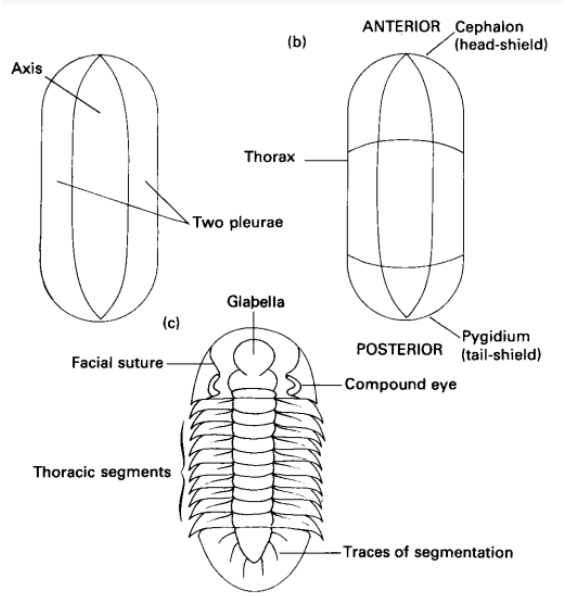
Figures taken from: Jones, Duck and Weyers' Practical skills in environmental science, 2000, Prentice Hall

Exercise 1 Drawing a trilobite:

We will do this altogether. The trilobite we are going to draw *is Paradoxides*.



This specimen is 24cm long



Exercise 2

In your groups the next trilobite to draw is *Calymene*

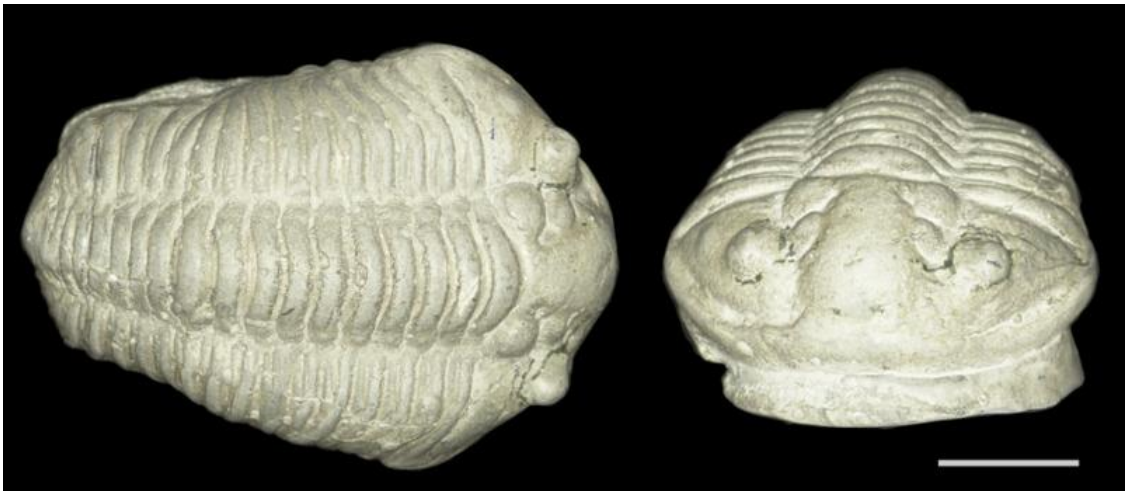


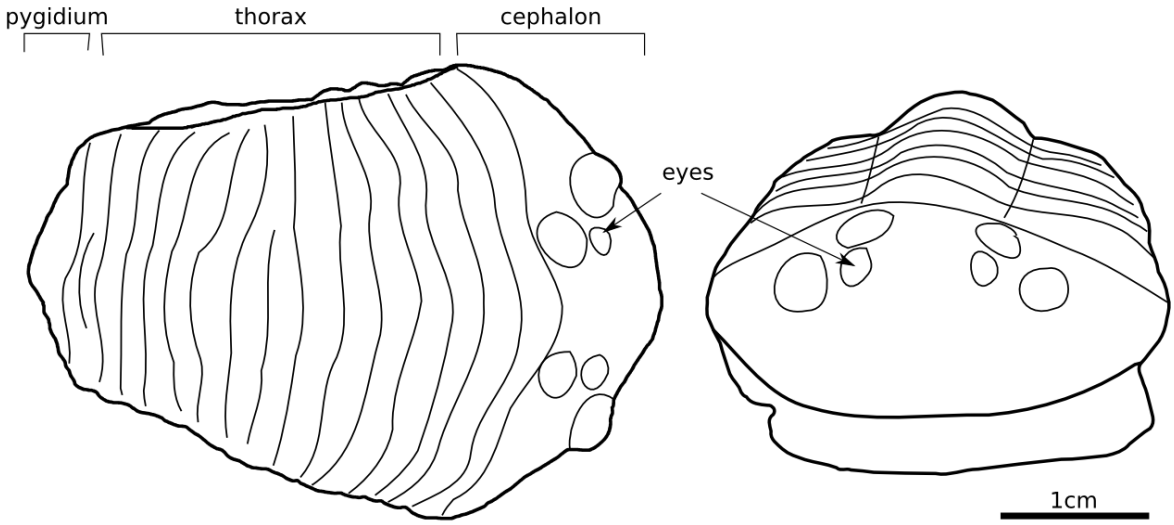
1cm

This image and drawing are taken from the University of Cambridge 1st year collection web site:

https://wserv3.esc.cam.ac.uk/p1acollections/

What do you think about the quality of this drawing of a *Calymene* fossil – how could you improve it? Draw your own version incorporating the improvements.





A good website all about trilobites, with lots of good diagrams:

https://www.trilobites.info/

https://www.trilobites.info/drawing.htm

If you are interested in learning some more about drawing fossils then here is some further reading:

Nield, Ted. *Drawing and Understanding Fossils : a Theoretical and Practical Guide for Beginners, with Self-Assessment* . Pergamon Press, 1987.

A practical manual for the student of palaeontology, giving a grounding in theory, in addition to teaching graphical skills needed to make clear, representative and pleasing drawings of fossil specimens. As most early practical work in palaeontology is graphical the author has sought to teach the basics of graphic art as relevant to each fossil group. Puzzles, exercises and experiments are included, also self-assessment tests to allow students to check their progress.

Available as an online book through the Main Library